How To Learn Math for Machine Learning FAST (Even With Zero Math Background)_transcript_20260201_135206

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Language Mode: Auto-detect (Mixed)

Detected: en (99.9%)

Transcribed: 2026-02-01 13:54:13

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Do you want to become a data scientist or machine learning engineer, but you feel intimidated by all ; the math involved? I get it, I've been there. I dropped out of high school after 10th grade, ; so I never learned any math beyond trigonometry in school. When I started my journey into machine ; learning, I didn't even know what a derivative was. Fast forward to today, and I'm an applied ; scientist at Amazon, and I feel pretty confident in my math skills. I've picked up the necessary ; math along the way using free resources and self-directed learning. Today I'm going to walk ; you through some of my favorite books, courses, and YouTube channels that helped me to get where ; I am today, and I'll also share some study tips on how to study effectively and not waste your ; time struggling and being bored. So let's get to it. First, let's address a common question. Do you ; even really need to know the math to work in machine learning? The short answer is, it depends ; on what you want to do. For research-heavy roles where you're creating new machine learning ; algorithms, then yes, you obviously need to know the math. But if you're asking yourself if you ; need to learn math, chances are that's not the kind of job you're looking for. For practitioners, ; most of us in the industry, you can be totally competent without knowing all the underlying ; details. ; At this point, libraries like NumPy, Scikit Learn, and TensorFlow handle most of the heavy ; lifting for you. You don't need to know the math behind gradient descent to deploy a model to ; production. If you're a beginner trying to get into machine learning, in my opinion it's not ; strategic to spend a bunch of time memorizing formulas or studying linear algebra. You should ; be spending that time building things. Train a simple model, explore your data, build a pipeline ; that predicts something fun. That said, there are moments where knowing the math really helps. ; Here are a couple

of examples. So imagine you're training a model, and it's just not converging. ; If
you understand concepts like gradients and optimization functions, you'll know whether to ; adjust your
learning rate, try a different optimizer, or tweak your data processing. ; Or let's say
you're running a linear regression, and you're interpreting the coefficients. Without ; math knowledge, you
might miss problems like multicolinearity, which makes the coefficients ; unreliable. Then you might make
incorrect conclusions from the data, and cost the company ; millions and lose your job.
Just kidding. Kind of. We do really need to be careful when making ; business
decisions from the models that we build. So while you can, and should, get started
without ; deep math knowledge, you should be careful when making business decisions from the
models that we ; build. It's definitely reasonable to build your comfort with math over
time. Once you're hands-on, ; you'll start encountering problems that naturally push you to learn
more. When you need to debug or ; explain your results, that's when the math
is going to start to click because it's connected ; to something real. So seriously,
don't let fear of math stop you from starting. You don't need to ; learn
it all up front to make progress. Get your hands dirty with the tools, build
your portfolio, ; and let math grow as a skill alongside your practical knowledge. Alright,
now let's talk ; about what to learn when you're building your math foundation for
machine learning jobs. ; First, linear algebra. ; Linear algebra is fundamental for machine learning,
especially for deep learning. Many ; models rely on representing data and computations as matrices
and vectors. Here's what to prioritize. ; First, matrices and vectors. Think of matrices as
grids of numbers and vectors as lists of numbers. ; Data is often stored this
way, and operations like addition, multiplication, and dot products ; are really central to how
models process that information. Determinants and inverses. ; Determinants tell you whether a matrix can
be inverted, which is used in optimization problems ; and solving systems of linear equations.
Eigenvalues and eigenvectors. ; These are key to understanding variance in data and are foundational
of techniques like ; principal component analysis, which can help to reduce the dimensionality in
datasets. ; Lastly, matrix decomposition. Methods like singular value decomposition are used in
recommendation ;
systems, dimensionality reduction, and data compression. ; Now we're onto basic calculus. Calculus is core
to understanding how models learn from ; data. But we don't need to worry about
solving super complex problems, it's just about grasping ; a couple of key ideas. ;
First, derivatives and gradients. Derivatives measure how things change and gradients, which ; are
multidimensional

derivatives, are what we call key ideas. Most of the time, they're ; power optimization
algorithms like gradient descent these help models adjust their parameters ; to minimize error the
chain rule is central to neural networks it's how back propagation works ; which is
the process of figuring out how much each weight in the network contributes to the
overall ; error so the model can learn effectively lastly optimization basics concepts like local
versus ; global minima saddle points and convexity are important to understand why some models
get stuck ; and others manage to find the best solutions lastly statistics and probability
statistics ; and probability are the bread and butter of understanding data while they're kind
of more ; associated with data science there's definitely a lot of value for machine
learning as well ; here's what you need to know distributions get familiar with common
distributions like the normal ; binomial and uniform the normal distribution in particular pops up
everywhere in data science and ; machine learning variance and covariance variance tells you how
spread out your data is while ; covariance shows you how two variables relate these
concepts are really important for feature ; selection and understanding your data structure bayes theorem
well it has kind of a silly maybe ; intimidating name bayes theorem is a
pretty simple but powerful tool for probabilistic ; reasoning it's foundational for algorithms like naive
bayes big surprise which is used for things ; like spam detection as well as
for bayesian optimization for hyperparameter tuning you'll ; also want to understand maximum likelihood estimation
which helps to estimate model ; parameters by finding values that maximize the likelihood of
your data it's a really fundamental ; concept especially in algorithms like logistic regression finally
sampling and conditional ; probability sampling lets you work with subsets of data efficiently and
conditional probability is ; essential for understanding the relationship between events especially in bayesian models
; it's definitely not an exhaustive list but i think it's a good overview of
the common concepts ; you'll need to know to do a good job as a
data scientist who's focused on machine learning or ; as a machine learning engineer next
up i'll share the best resources to learn these concepts without ; it being stressful
or overwhelming and hopefully also not too boring personally i would highly ; recommend starting
with a visual and intuitive understanding of the key concepts before you start ; reading
really difficult textbooks and trying to solve equations for linear algebra and calculus ; i
cannot speak highly enough about three blue one brown's essence of linear algebra and essence
of ; calculus series ; so i'm going to start with the first one and

i'm going to give you a little bit of a ; so first of all
i want to give a really solid introduction to what is actually being measured ;
and manipulated when we use these mathematical approaches more importantly they kind of show ;
the beauty in it it's kind of strange to say that math videos could be
inspirational in a way but ; these ones are for statistics and probability i am
a huge fan of statquest his videos are clear ; engaging and just a joy
to watch statquest has playlists with overviews on core stats and machine ; learning concepts
so i'd recommend starting there once you have a visual intuition you can start ;
working through more structured books and courses there are a lot of great options here
as well let's ; go through a few that i personally use to learn i
completed the mathematics for machine learning ; specialization from imperial college london on coursera when
i was just starting out the ; specialization is divided into three courses linear algebra
multivariate calculus and a ; last one on principal component analysis the courses are well
structured and include a mix ; of video lectures quizzes and programming assignments in python
i found the course to be a ; little bit challenging as a beginner but
it was a really good overview and i did manage to pass ; everything with
a little bit of effort deeplearning.ai also recently released a math for ; machine learning
specialization on coursera this specialization has courses on linear algebra and ; calculus but instead
of pca the final course focuses on stats and probability i'm personally ; working through
this specialization right now and overall i'm finding it to be another really great ;
option each module starts with a nice motivation for how the math connects to an
applied machine ; learning concept it has coding exercises in python and some neat 3d
tools to mess around with to get ; understanding of the concepts as well if
you prefer learning from books i have some suggestions ; there too first up if
you like anime or nerdy stuff boy oh boy do i have a recommendation for
you ; did you know that they have manga math books so i've read the
manga guide to calculus manga guide ; to linear algebra and manga guide to statistics
they also have a ton more they have someone like ; physics i think and
some other engineering concepts so they're super super fun i can't say the ; instructional
quality is like world class or anything but they're cute and engaging and they ;
are really fun to learn and i think they're great for learning a math book
the next level up would ; be real math books these are some of the

best the mathematics for machine learning ebook by ; dyson roth and colleagues is a
great comprehensive resource and it's available for free for personal ; use it covers key
topics we've already discussed like linear algebra calculus probability and ; optimization with a focus
on how these concepts apply to machine learning algorithms it's ; relatively beginner friendly and
is generally regarded as one of the best books for learning ; this material next
practical statistics for data scientists is another well-loved resource that ; are now before we
actually start studying i think it's important to spend a little bit of time ;
thinking really deeply about why you even want to do this personally i find that
if i'm studying ; just because i feel like i should or because it's some
arbitrary assignment i get distracted easily ; and i don't actually end up retaining that
much of the information instead i try to connect to ; some kind of a
deeper motivation if possible personally right now i have a really basic ; motivation i
want to do well in my job because i want to earn a lot of
money so that i can take ; care of everyone i love i have these
great resources to learn from that weren't even ; available a couple of years ago
and i have this opportunity to push myself and make sure that ; everyone i
love is safe and cared for now and in the future so this isn't to
put extra pressure ; on myself but actually just a way that works for me
to kind of get excited that i have this ; opportunity to learn and grow
and hopefully help others along the way your motivation might be ; totally different but
whatever it is try to tie this work to a larger goal in terms of
optimizing ; your study time i found that one of the most effective methods is
writing notes in my own words ; don't just copy definitions or formulas ; take
time to summarize concepts as if you were explaining them to someone else or really
to ; future you for example if you're learning about derivatives you might write a
derivative measures ; how a function changes given a tiny nudge to its input this
forces you to actively process ; the material relatedly when it comes to math formulas
don't just stare at them i know it's ; easy to do it's really helpful
to translate them into plain english or whatever spoken language you ; prefer for instance
take the equation y equals mx plus b you might describe m as the slope
that ; shows how steep the line is and b as the slope that shows
how steep the line is and b as the ; point where the line crosses
the y-axis so the final formula might be the value of y the output ;

is determined by taking the slope m multiplying it by x the input and then
adding b the starting ; point where the line intersects the y-axis you can even
use your notes as like a personal blog ; kind of writing short posts about
what you've learned is a really solid way to clarify your ; understanding and teaching
others even if nobody reads it solidifies the material in your own mind ; plus
sharing your posts on medium or linkedin not only potentially helps others but also ;
allows you to build kind of a portfolio showcasing your learning journey also trust me
when it's ; interview time you're going to be really happy that you have these
notes i use my own study ; notes all the time the next piece of
advice i have might not be super duper fun but i also recommend ; not
using just one resource personally i've had a lot of success from taking many different
courses ; and kind of just throwing all my notes together at first then i'll
write a blog like i was just ; talking about that summarizes all of my
learnings there are a couple of advantages to this approach ; first repetition helps you
retain things if i see a concept multiple ; times explained from multiple angles and
much more likely to actually get what's going on ; and remember it for longer
than a day plus not only do i see the information presented ; to me
multiple times i'm writing the concepts out in my own words multiple times including ;
that final time where i synthesize it all and get ready to share with others
so i have to be ; really confident that i actually got it by the
end if i'm going to be putting it out publicly ; let me show you
an example here are three different explanations for the concept of rank ; and linear
algebra starting with three blue one brown you can see this visual intuition which is
; a little bit more complicated than the one i just showed you but you
know it lacks a little detail ; then we can look at how rank was
presented in deeplearning.ai which has some more detail but ; is still presented in a
really simple way finally we can go to the mathematics for machine learning ; ebook
to get an even more formalized definition with additional detail each one of these builds
; on the other and adds to my store of knowledge finally once you've built
that foundation and ; get to the level of math where you can actually use
it for stuff i really recommend coding ; concepts from scratch if you can code
gradient descent or logistic regression using just numpy ; you're off to a really strong
start while i know at this point you are ; super excited to start learning

math i do want to just circle back to the important ; fact that if
you're a beginner trying to get your first job in my opinion math should not
; be the first thing you prioritize it's really unlikely that your math skills are
going to be ; the thing that gets you a job as a data scientist
or machine learning engineer instead prioritize ; gaining hands-on experience by working on projects and
actually building stuff employers are far more ; interested in seeing what you can do
with the tools and knowledge you already have than how ; many formulas you've memorized
as you encounter challenges in your work you'll naturally be ; the math behind the
algorithms remember math is a tool to help you succeed and shouldn't be a ;
barrier to getting started now if you're curious about what else you need to know
to become a data ; scientist i have a really comprehensive video that lays out
the entire roadmap from absolute ; beginner to your first job including skills to learn
project examples how to get experience ; preparing for interviews and more definitely check that
out next

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TIMESTAMPS:

[00:00 - 00:04]
Do you want to become a data scientist or machine learning engineer, but you feel
intimidated by all

[00:04 - 00:08]
the math involved? I get it, I've been there. I dropped out of high school
after 10th grade,

[00:08 - 00:13]
so I never learned any math beyond trigonometry in school. When I started my journey
into machine

[00:13 - 00:16]
learning, I didn't even know what a derivative was. Fast forward to today, and I'm
an applied

[00:16 - 00:20]
scientist at Amazon, and I feel pretty confident in my math skills. I've picked up
the necessary

[00:20 - 00:25]
math along the way using free resources and self-directed learning. Today I'm going to walk

[00:25 - 00:28]
you through some of my favorite books, courses, and YouTube channels that helped me to
get where

[00:28 - 00:32]
I am today, and I'll also share some study tips on how to study effectively
and not waste your

[00:32 - 00:38]
time struggling and being bored. So let's get to it. First, let's address a common
question. Do you

[00:38 - 00:41]
even really need to know the math to work in machine learning? The short answer
is, it depends

[00:41 - 00:45]
on what you want to do. For research-heavy roles where you're creating new machine learning

[00:45 - 00:49]
algorithms, then yes, you obviously need to know the math. But if you're asking yourself
if you

[00:49 - 00:53]
need to learn math, chances are that's not the kind of job you're looking for.

For practitioners,

[00:53 - 00:57]
most of us in the industry, you can be totally competent without knowing all the
underlying

[00:57 - 00:58]
details.

[00:58 - 01:03]
At this point, libraries like NumPy, Scikit Learn, and TensorFlow handle most of the heavy

[01:03 - 01:07]
lifting for you. You don't need to know the math behind gradient descent to deploy
a model to

[01:07 - 01:11]
production. If you're a beginner trying to get into machine learning, in my opinion it's
not

[01:11 - 01:15]
strategic to spend a bunch of time memorizing formulas or studying linear algebra. You should

[01:15 - 01:20]
be spending that time building things. Train a simple model, explore your data, build a
pipeline

[01:20 - 01:24]
that predicts something fun. That said, there are moments where knowing the math really helps.

[01:24 - 01:28]
Here are a couple of examples. So imagine you're training a model, and it's just
not converging.

[01:28 - 01:32]
If you understand concepts like gradients and optimization functions, you'll know whether to

[01:32 - 01:36]
adjust your learning rate, try a different optimizer, or tweak your data processing.

[01:36 - 01:40]
Or let's say you're running a linear regression, and you're interpreting the coefficients. Without

[01:40 - 01:44]
math knowledge, you might miss problems like multicolinearity, which makes the coefficients

[01:44 - 01:48]
unreliable. Then you might make incorrect conclusions from the data, and cost the company

[01:48 - 01:52]
millions and lose your job. Just kidding. Kind of. We do really need to be
careful when making

[01:52 - 01:57]
business decisions from the models that we build. So while you can, and should, get
started without

[01:57 - 01:58]
deep math knowledge, you should be careful when making business decisions from the models that
we

[01:58 - 02:01]
build. It's definitely reasonable to build your comfort with math over time. Once you're hands-on,

[02:02 - 02:06]
you'll start encountering problems that naturally push you to learn more. When you need to

debug or

[02:06 - 02:09]
explain your results, that's when the math is going to start to click because it's
connected

[02:09 - 02:14]
to something real. So seriously, don't let fear of math stop you from starting. You
don't need to

[02:14 - 02:18]
learn it all up front to make progress. Get your hands dirty with the tools,
build your portfolio,

[02:18 - 02:23]
and let math grow as a skill alongside your practical knowledge. Alright, now let's talk

[02:23 - 02:25]
about what to learn when you're building your math foundation for machine learning jobs.

[02:26 - 02:27]
First, linear algebra.

[02:27 - 02:31]
Linear algebra is fundamental for machine learning, especially for deep learning. Many

[02:31 - 02:35]
models rely on representing data and computations as matrices and vectors. Here's what to prioritize.

[02:36 - 02:41]
First, matrices and vectors. Think of matrices as grids of numbers and vectors as lists
of numbers.

[02:41 - 02:45]

Data is often stored this way, and operations like addition, multiplication, and dot products

[02:45 - 02:49]

are really central to how models process that information. Determinants and inverses.

[02:49 - 02:53]

Determinants tell you whether a matrix can be inverted, which is used in optimization problems

[02:53 - 02:56]

and solving systems of linear equations. Eigenvalues and eigenvectors.

[02:57 - 03:00]

These are key to understanding variance in data and are foundational of techniques like

[03:00 - 03:04]

principal component analysis, which can help to reduce the dimensionality in datasets.

[03:04 - 03:09]

Lastly, matrix decomposition. Methods like singular value decomposition are used in recommendation

[03:09 - 03:11]

systems, dimensionality reduction, and data compression.

[03:12 - 03:15]

Now we're onto basic calculus. Calculus is core to understanding how models learn from

[03:15 - 03:19]

data. But we don't need to worry about solving super complex problems, it's just about

grasping

[03:19 - 03:20]

a couple of key ideas.

[03:21 - 03:25]

First, derivatives and gradients. Derivatives measure how things change and gradients, which

[03:25 - 03:27]

are multidimensional derivatives, are what we call key ideas. Most of the time, they're

[03:27 - 03:31]

power optimization algorithms like gradient descent these help models adjust their parameters

[03:31 - 03:35]

to minimize error the chain rule is central to neural networks it's how back propagation works

[03:35 - 03:39]

which is the process of figuring out how much each weight in the network contributes to the overall

[03:39 - 03:44]

error so the model can learn effectively lastly optimization basics concepts like local versus

[03:44 - 03:49]

global minima saddle points and convexity are important to understand why some models get stuck

[03:49 - 03:53]

and others manage to find the best solutions lastly statistics and probability statistics

[03:53 - 03:56]

and probability are the bread and butter of understanding data while they're kind of more

[03:56 - 03:59]

associated with data science there's definitely a lot of value for machine learning as well

[03:59 - 04:04]

here's what you need to know distributions get familiar with common distributions like the normal

[04:04 - 04:09]

binomial and uniform the normal distribution in particular pops up everywhere in data science and

[04:09 - 04:13]

machine learning variance and covariance variance tells you how spread out your data is while

[04:13 - 04:17]

covariance shows you how two variables relate these concepts are really important for feature

[04:17 - 04:22]

selection and understanding your data structure bayes theorem well it has kind of a silly
maybe

[04:22 - 04:26]

intimidating name bayes theorem is a pretty simple but powerful tool for probabilistic

[04:26 - 04:30]

reasoning it's foundational for algorithms like naive bayes big surprise which is used for things

[04:30 - 04:35]

like spam detection as well as for bayesian optimization for hyperparameter tuning you'll

[04:35 - 04:38]

also want to understand maximum likelihood estimation which helps to estimate model

[04:38 - 04:42]

parameters by finding values that maximize the likelihood of your data it's a really fundamental

[04:42 - 04:47]

concept especially in algorithms like logistic regression finally sampling and conditional

[04:47 - 04:51]
probability sampling lets you work with subsets of data efficiently and conditional probability is

[04:51 - 04:55]
essential for understanding the relationship between events especially in bayesian models

[04:56 - 04:59]
it's definitely not an exhaustive list but i think it's a good overview of the
common concepts

[04:59 - 05:03]
you'll need to know to do a good job as a data scientist who's focused
on machine learning or

[05:03 - 05:07]
as a machine learning engineer next up i'll share the best resources to learn these
concepts without

[05:07 - 05:11]
it being stressful or overwhelming and hopefully also not too boring personally i would highly

[05:11 - 05:16]
recommend starting with a visual and intuitive understanding of the key concepts before you start

[05:16 - 05:20]
reading really difficult textbooks and trying to solve equations for linear algebra and calculus

[05:20 - 05:24]
i cannot speak highly enough about three blue one brown's essence of linear algebra and
essence of

[05:24 - 05:25]

calculus series

[05:26 - 05:26]
so i'm going to start with the first one and i'm going to give you
a little bit of a

[05:26 - 05:29]
so first of all i want to give a really solid introduction to what is
actually being measured

[05:29 - 05:33]
and manipulated when we use these mathematical approaches more importantly they kind of show

[05:33 - 05:37]
the beauty in it it's kind of strange to say that math videos could be
inspirational in a way but

[05:37 - 05:42]
these ones are for statistics and probability i am a huge fan of statquest his
videos are clear

[05:42 - 05:47]
engaging and just a joy to watch statquest has playlists with overviews on core stats
and machine

[05:47 - 05:52]
learning concepts so i'd recommend starting there once you have a visual intuition you can
start

[05:52 - 05:57]
working through more structured books and courses there are a lot of great options here
as well let's

[05:57 - 06:00]
go through a few that i personally use to learn i completed the mathematics for
machine learning

[06:00 - 06:04]
specialization from imperial college london on coursera when i was just starting out the

[06:04 - 06:09]
specialization is divided into three courses linear algebra multivariate calculus and a

[06:09 - 06:13]
last one on principal component analysis the courses are well structured and include a mix

[06:13 - 06:17]
of video lectures quizzes and programming assignments in python i found the course to be
a

[06:17 - 06:21]
little bit challenging as a beginner but it was a really good overview and i
did manage to pass

[06:21 - 06:25]
everything with a little bit of effort deeplearning.ai also recently released a math for

[06:25 - 06:29]
machine learning specialization on coursera this specialization has courses on linear algebra and

[06:29 - 06:33]
calculus but instead of pca the final course focuses on stats and probability i'm personally

[06:33 - 06:37]
working through this specialization right now and overall i'm finding it to be another really
great

[06:37 - 06:41]
option each module starts with a nice motivation for how the math connects to an
applied machine

[06:41 - 06:46]
learning concept it has coding exercises in python and some neat 3d tools to mess
around with to get

[06:47 - 06:51]
understanding of the concepts as well if you prefer learning from books i have some
suggestions

[06:51 - 06:56]
there too first up if you like anime or nerdy stuff boy oh boy do
i have a recommendation for you

[06:57 - 07:02]
did you know that they have manga math books so i've read the manga guide
to calculus manga guide

[07:02 - 07:06]
to linear algebra and manga guide to statistics they also have a ton more they
have someone like

[07:06 - 07:11]
physics i think and some other engineering concepts so they're super super fun i can't
say the

[07:11 - 07:16]
instructional quality is like world class or anything but they're cute and engaging and they

[07:17 - 07:19]

are really fun to learn and i think they're great for learning a math book
the next level up would

[07:19 - 07:24]
be real math books these are some of the best the mathematics for machine learning
ebook by

[07:24 - 07:28]
dyson roth and colleagues is a great comprehensive resource and it's available for free for
personal

[07:28 - 07:33]
use it covers key topics we've already discussed like linear algebra calculus probability and

[07:33 - 07:37]
optimization with a focus on how these concepts apply to machine learning algorithms it's

[07:37 - 07:41]
relatively beginner friendly and is generally regarded as one of the best books for learning

[07:41 - 07:46]
this material next practical statistics for data scientists is another well-loved resource that

[07:47 - 07:51]
are now before we actually start studying i think it's important to spend a little
bit of time

[07:51 - 07:56]
thinking really deeply about why you even want to do this personally i find that
if i'm studying

[07:56 - 08:01]
just because i feel like i should or because it's some arbitrary assignment i get

distracted easily

[08:01 - 08:05]
and i don't actually end up retaining that much of the information instead i try
to connect to

[08:05 - 08:09]
some kind of a deeper motivation if possible personally right now i have a really
basic

[08:09 - 08:13]
motivation i want to do well in my job because i want to earn a
lot of money so that i can take

[08:13 - 08:17]
care of everyone i love i have these great resources to learn from that weren't
even

[08:17 - 08:22]
available a couple of years ago and i have this opportunity to push myself and
make sure that

[08:22 - 08:27]
everyone i love is safe and cared for now and in the future so this
isn't to put extra pressure

[08:27 - 08:31]
on myself but actually just a way that works for me to kind of get
excited that i have this

[08:31 - 08:35]
opportunity to learn and grow and hopefully help others along the way your motivation might
be

[08:35 - 08:40]
totally different but whatever it is try to tie this work to a larger goal
in terms of optimizing

[08:40 - 08:45]
your study time i found that one of the most effective methods is writing notes
in my own words

[08:45 - 08:47]
don't just copy definitions or formulas

[08:47 - 08:51]
take time to summarize concepts as if you were explaining them to someone else or
really to

[08:51 - 08:56]
future you for example if you're learning about derivatives you might write a derivative measures

[08:56 - 09:01]
how a function changes given a tiny nudge to its input this forces you to
actively process

[09:01 - 09:05]
the material relatedly when it comes to math formulas don't just stare at them i
know it's

[09:05 - 09:09]
easy to do it's really helpful to translate them into plain english or whatever spoken
language you

[09:09 - 09:15]
prefer for instance take the equation y equals mx plus b you might describe m

as the slope that

[09:15 - 09:17]
shows how steep the line is and b as the slope that shows how steep
the line is and b as the

[09:17 - 09:23]
point where the line crosses the y-axis so the final formula might be the value
of y the output

[09:23 - 09:28]
is determined by taking the slope m multiplying it by x the input and then
adding b the starting

[09:28 - 09:33]
point where the line intersects the y-axis you can even use your notes as like
a personal blog

[09:33 - 09:37]
kind of writing short posts about what you've learned is a really solid way to
clarify your

[09:37 - 09:42]
understanding and teaching others even if nobody reads it solidifies the material in your own
mind

[09:42 - 09:47]
plus sharing your posts on medium or linkedin not only potentially helps others but also

[09:47 - 09:52]
allows you to build kind of a portfolio showcasing your learning journey also trust me
when it's

[09:52 - 09:55]
interview time you're going to be really happy that you have these notes i use
my own study

[09:55 - 10:00]
notes all the time the next piece of advice i have might not be super
duper fun but i also recommend

[10:00 - 10:05]
not using just one resource personally i've had a lot of success from taking many
different courses

[10:05 - 10:09]
and kind of just throwing all my notes together at first then i'll write a
blog like i was just

[10:09 - 10:13]
talking about that summarizes all of my learnings there are a couple of advantages to
this approach

[10:13 - 10:17]
first repetition helps you retain things if i see a concept multiple

[10:17 - 10:21]
times explained from multiple angles and much more likely to actually get what's going on

[10:21 - 10:25]
and remember it for longer than a day plus not only do i see the
information presented

[10:25 - 10:30]
to me multiple times i'm writing the concepts out in my own words multiple times
including

[10:30 - 10:34]
that final time where i synthesize it all and get ready to share with others
so i have to be

[10:34 - 10:37]
really confident that i actually got it by the end if i'm going to be
putting it out publicly

[10:37 - 10:42]
let me show you an example here are three different explanations for the concept of
rank

[10:42 - 10:46]
and linear algebra starting with three blue one brown you can see this visual intuition
which is

[10:47 - 10:49]
a little bit more complicated than the one i just showed you but you know
it lacks a little detail

[10:49 - 10:54]
then we can look at how rank was presented in deeplearning.ai which has some more
detail but

[10:54 - 10:58]
is still presented in a really simple way finally we can go to the mathematics
for machine learning

[10:58 - 11:02]
ebook to get an even more formalized definition with additional detail each one of these
builds

[11:02 - 11:06]
on the other and adds to my store of knowledge finally once you've built that
foundation and

[11:06 - 11:10]
get to the level of math where you can actually use it for stuff i
really recommend coding

[11:10 - 11:15]
concepts from scratch if you can code gradient descent or logistic regression using just numpy

[11:15 - 11:17]
you're off to a really strong start while i know at this point you are

[11:17 - 11:21]
super excited to start learning math i do want to just circle back to the
important

[11:21 - 11:26]
fact that if you're a beginner trying to get your first job in my opinion
math should not

[11:26 - 11:30]
be the first thing you prioritize it's really unlikely that your math skills are going
to be

[11:30 - 11:34]
the thing that gets you a job as a data scientist or machine learning engineer
instead prioritize

[11:34 - 11:39]
gaining hands-on experience by working on projects and actually building stuff employers are far more

[11:39 - 11:43]
interested in seeing what you can do with the tools and knowledge you already have
than how

[11:43 - 11:47]
many formulas you've memorized as you encounter challenges in your work you'll naturally be

[11:47 - 11:51]
the math behind the algorithms remember math is a tool to help you succeed and
shouldn't be a

[11:51 - 11:55]
barrier to getting started now if you're curious about what else you need to know
to become a data

[11:55 - 11:59]
scientist i have a really comprehensive video that lays out the entire roadmap from absolute

[11:59 - 12:04]
beginner to your first job including skills to learn project examples how to get experience

[12:04 - 12:08]
preparing for interviews and more definitely check that out next

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TRANSLATION (ID):

Apakah Anda ingin menjadi insinyur ilmuwan data (data scientist) atau pembelajaran mesin (machine learning), tetapi
Anda merasa terintimidasi oleh semua orang matematika yang terlibat? Saya mengerti, saya pernah ke sana.
Saya putus sekolah setelah kelas 10, jadi saya tidak pernah belajar matematika selain trigonometri di
sekolah. Saat saya memulai perjalanan saya ke mesin belajar, saya bahkan tidak tahu apa itu

turunan. Maju cepat ke hari ini, dan saya sudah melamar ilmuwan di Amazon, dan saya merasa cukup percaya diri dengan kemampuan matematika saya. Aku sudah mengambil yang diperlukan matematika sepanjang jalan menggunakan sumber daya gratis dan pembelajaran mandiri. Hari ini aku akan berjalan Anda melalui beberapa buku, kursus, dan saluran YouTube favorit saya yang membantu saya mencapai tujuan Saya hari ini, dan saya juga akan berbagi beberapa tips belajar tentang cara belajar yang efektif dan tidak menyia-nyiakan waktu Anda waktu berjuang dan merasa bosan. Jadi mari kita mulai. Pertama, mari kita jawab pertanyaan umum. Apakah kamu bahkan benar-benar perlu mengetahui matematika untuk bekerja di pembelajaran mesin (machine learning)? Jawaban singkatnya adalah, itu tergantung pada apa yang ingin Anda lakukan. Untuk peran yang banyak melakukan penelitian saat Anda membuat pembelajaran mesin (machine learning) baru algoritma (algorithm)s, maka ya, Anda jelas perlu mengetahui matematika. Tetapi jika Anda bertanya pada diri sendiri apakah Anda perlu belajar matematika, kemungkinan besar itu bukan pekerjaan yang Anda cari. Bagi praktisi, kebanyakan dari kita di industri (industry), Anda bisa menjadi benar-benar kompeten tanpa mengetahui semua hal yang mendasarinya detail. Pada titik ini, perpustakaan seperti NumPy, Scikit Learn, dan tensor (tensor)Flow menangani sebagian besar tugas berat mengangkat untukmu. Anda tidak perlu mengetahui matematika di balik penurunan gradien (gradient descent) untuk menerapkan model (model) ke produksi. Jika anda seorang pemula yang mencoba masuk ke pembelajaran mesin (machine learning), menurut saya tidak strategis untuk menghabiskan banyak waktu menghafal rumus atau mempelajari aljabar linier. Anda harus menghabiskan waktu itu untuk membangun sesuatu. Latih model (model) sederhana, jelajahi data Anda, buat pipa (pipeline) yang memprediksi sesuatu yang menyenangkan. Meskipun demikian, ada saat-saat di mana mengetahui matematika sangat membantu. Berikut beberapa contohnya. Jadi bayangkan Anda pelatihan (training) dan model (model), dan itu tidak menyatu. Jika Anda memahami konsep seperti gradien dan fungsi pengoptimalan, Anda akan mengetahui apakah Anda perlu memahaminya sesuaikan laju pembelajaran (learning rate) Anda,

coba pengoptimal
lain, atau sesuaikan proses (process) data Anda. Atau katakanlah Anda menjalankan regresi linier (linear regression),
dan Anda menafsirkan koefisiennya. Tanpa pengetahuan matematika, Anda mungkin melewatkan soal seperti multikolinearitas, yang menghasilkan
koefisien tidak dapat diandalkan. Kemudian Anda mungkin membuat kesimpulan yang salah dari data tersebut, dan
merugikan perusahaan jutaan dan kehilangan pekerjaan Anda. Cuma bercanda. Agak. Kita memang harus berhati-hati saat
membuatnya keputusan bisnis dari model (model)s yang kami bangun. Jadi selagi Anda bisa, dan harus,
memulai tanpa pengetahuan matematika yang mendalam, Anda harus berhati-hati saat mengambil keputusan bisnis dari model
(model)s yang kita miliki membangun. Sangat masuk akal untuk membangun kenyamanan Anda dengan matematika seiring
berjalannya waktu. Setelah Anda siap, Anda akan mulai menghadapi masalah yang secara alami mendorong Anda
untuk belajar lebih banyak. Saat Anda perlu debug (debug) atau jelaskan hasilmu, saat itulah matematika
akan mulai diklik karena terhubung untuk sesuatu yang nyata. Serius, jangan biarkan rasa takut terhadap
matematika menghentikan Anda untuk memulai. Anda tidak perlu melakukannya pelajari semuanya terlebih dahulu untuk membuat
kemajuan. Jadikan tangan Anda kotor dengan alat (tool)s, buat porta (port)folio Anda, dan biarkan matematika
berkembang sebagai keterampilan seiring dengan pengetahuan praktis Anda. Baiklah, sekarang mari kita bicara tentang apa
yang harus dipelajari saat Anda membangun landasan matematika untuk pekerjaan pembelajaran mesin (machine learning). Pertama,
aljabar linier. Aljabar linier merupakan dasar untuk pembelajaran mesin (machine learning), terutama untuk pembelajaran mendalam
(deep learning). Banyak model (model)s mengandalkan representasi data dan komputasi sebagai matriks dan vektor (vector)s.
Inilah yang harus diprioritaskan. Pertama, matriks dan vektor (vector)s. Bayangkan matriks sebagai kisi-kisi angka dan
vektor (vector)s sebagai daftar angka. Data sering kali disimpan dengan cara ini, dan operasi seperti

penjumlahan, perkalian, dan perkalian titik sangat penting dalam cara model (model)s proses (process) informasi tersebut.

Penentu dan invers. Penentu memberi tahu Anda apakah suatu matriks dapat dibalik, yang digunakan dalam masalah optimasi dan menyelesaikan sistem (system)s persamaan linier. Nilai eigen dan eigenvektor (vector)s. Ini adalah

kunci untuk memahami varians (variance) dalam data dan merupakan dasar dari teknologi (tech)niques seperti analisis

komponen utama (principal component analysis), yang dapat membantu mengurangi dimensi dalam dataset (dataset)s. Terakhir, dekomposisi

matriks. Metode seperti dekomposisi nilai tunggal digunakan dalam rekomendasi sistem (system)s, reduksi dimensi, dan data

mampatkan (compress)ion. Sekarang kita masuk ke kalkulus dasar. Kalkulus adalah inti untuk memahami bagaimana model

(model)s belajar data. Namun kita tidak perlu khawatir untuk memecahkan masalah yang super rumit, yang penting adalah pemahaman beberapa ide kunci. Pertama, turunan dan gradien. Derivatif mengukur bagaimana sesuatu berubah

dan gradien, yang mana adalah turunan multidimensi, itulah yang kami sebut ide-ide kunci. Seringkali memang

demikian pengoptimalan daya algoritma (algorithm) seperti penurunan gradien (gradient descent) ini membantu model (model) menyesuaikan

parameter (parameter)nya untuk meminimalkan galat (error) aturan rantai sangat penting bagi jaringan saraf (neural network)s

begitulah cara kerja propagasi balik yang merupakan proses (process) untuk mencari tahu seberapa besar kontribusi

masing-masing bobot (weight) di jaringan (network) terhadap keseluruhan galat (error) sehingga model (model) dapat mempelajari

secara efektif dan terakhir konsep dasar pengoptimalan seperti lokal versus titik pelana minimum global dan konveksitasnya penting untuk memahami mengapa beberapa model (model) terjebak dan lainnya berhasil menemukan solusi terbaik,

terakhir statistik dan probabilitas (probability) statistik dan probabilitas (probability) adalah inti dari data pemahaman, sementara

keduanya lebih dari itu dikaitkan dengan sains data (data science) pasti ada banyak nilai untuk

pembelajaran mesin (machine learning) juga inilah yang perlu anda ketahui tentang distribusi. Biasakan diri

dengan
distribusi umum seperti biasanya binomial dan seragam distribusi normal khususnya muncul di mana-mana di sains
data (data science) dan pembelajaran mesin (machine learning) varians (variance) dan covarians (variance) varians (variance)
memberi tahu Anda seberapa tersebar data Anda saat covarians (variance) menunjukkan kepada Anda bagaimana dua
variabel (variable) menghubungkan konsep-konsep ini yang sangat penting untuk fitur memilih dan memahami teorema bayes
struktur data Anda dengan baik mungkin agak konyol teorema nama bayes yang mengintimidasi adalah alat
(tool) yang cukup sederhana namun kuat untuk probabilistik alasan itu dasar untuk algoritma (algorithm) seperti
naive bayes (naive bayes) kejutan besar yang digunakan untuk berbagai hal seperti deteksi spam (spam)
serta untuk optimasi bayesian (bayesian) untuk penyetelan hyperparameter (hyperparameter tuning) Anda akan juga ingin memahami
estimasi kemungkinan maksimum yang membantu memperkirakan model (model) parameter (parameter)s dengan menemukan nilai yang memaksimalkan
kemungkinan data Anda, itu sangat mendasar konsep terutama di algoritma (algorithm)s seperti regresi logistik (logistic
regression) akhirnya sampling dan bersyarat Pengambilan sampel probabilitas (probability) memungkinkan Anda bekerja dengan subkumpulan data
secara efisien dan bersyarat probabilitas (probability) penting untuk memahami hubungan antar peristiwa terutama di bayesian
(bayesian) model (model)s ini jelas bukan daftar yang lengkap tetapi menurut saya ini adalah kumpulan
tampilan (view) yang bagus dari konsep-konsep umum Anda harus tahu untuk melakukan pekerjaan dengan baik
sebagai ilmuwan data (data scientist) yang fokus pada pembelajaran mesin (machine learning) atau sebagai insinyur
pembelajaran mesin (machine learning) selanjutnya saya akan membagikan sumber daya terbaik untuk mempelajari konsep-konsep ini
tanpanya itu membuat stres atau membebani dan mudah-mudahan juga tidak terlalu membosankan, secara pribadi saya
akan sangat melakukannya merekomendasikan memulai dengan visual (visual) dan pemahaman intuitif

tentang konsep-konsep utama sebelum
Anda memulai membaca buku teks (text) yang sangat sulit dan mencoba menyelesaikan persamaan untuk aljabar
linier dan kalkulus Saya tidak dapat berbicara cukup banyak tentang esensi aljabar linier tiga biru
satu coklat dan esensinya seri kalkulus jadi saya akan mulai dengan yang pertama dan saya
akan memberi Anda sedikit jadi pertama-tama saya ingin memberikan pengenalan yang solid tentang apa yang
sebenarnya diukur dan dimanipulasi ketika kita menggunakan pendekatan matematika ini secara lebih cepat, hal itu
terlihat jelas keindahan di dalamnya agak aneh untuk mengatakan bahwa matematika video (video)s bisa menjadi
inspirasi dalam beberapa hal, tetapi ini untuk statistik dan probabilitas (probability) saya penggemar berat statquest
video (video)nya jelas menarik dan menyenangkan untuk menonton statquest memiliki daftar putar dengan lebih daritampilan
(view)s pada statistik inti dan mesin mempelajari konsep jadi saya sarankan memulai dari sana setelah
Anda memiliki intuisi visual (visual), Anda dapat memulai bekerja melalui buku dan kursus yang lebih
terstruktur, ada banyak pilihan bagus di sini juga, mari lihat beberapa yang saya pribadi gunakan
untuk mempelajarinya. Saya menyelesaikan matematika untuk pembelajaran mesin (machine learning) spesialisasi dari imperial college london
di coursera ketika saya baru memulai peminatan dibagi menjadi tiga mata kuliah aljabar linier kalkulus
multivariat dan a yang terakhir di analisis komponen utama (principal component analysis) kursusnya terstruktur dengan
baik dan mencakup campuran dari video (video) kuis kuliah dan program (program)ming tugas dengan python
saya menemukan kursus ini sebagai sedikit menantang sebagai pemula tetapi itu adalah overtampilan (view) yang
sangat bagus dan saya berhasil lulus semuanya dengan sedikit usaha deeplearning.ai juga baru-baru ini merilis
matematika untuk pembelajaran mesin (machine learning) spesialisasi kursusra spesialisasi ini memiliki kursus aljabar linier dan
kalkulus tetapi alih-alih pca, kursus terakhir berfokus pada statistik dan probabilitas (probability) menurut saya pribadi

mengerjakan spesialisasi ini sekarang dan secara keseluruhan menurut saya ini adalah spesialisasi lain yang sangat
hebat opsi setiap modul dimulai dengan motivasi yang bagus tentang bagaimana matematika terhubung ke mesin
yang diterapkan konsep pembelajarannya memiliki latihan pengkodean dengan python dan beberapa alat (tool) 3d yang
rapi untuk dipusingkan pemahaman konsep juga jika Anda lebih suka belajar dari buku saya punya
beberapa saran di sana juga yang pertama jika Anda suka anime atau hal-hal kutu buku
ya ampun, apakah saya punya rekomendasi untuk Anda tahukah kamu kalau mereka punya buku matematika
manga jadi aku sudah membaca manga panduan (guide) hingga manga kalkulus panduan (guide) hingga aljabar
linier dan manga panduan (guide) hingga statistik, mereka juga punya lebih banyak lagi, mereka punya
seseorang seperti fisika menurut saya dan beberapa konsep teknik lainnya jadi sangat menyenangkan, saya tidak
bisa mengatakannya kualitas pengajarannya seperti kelas dunia atau apa pun, tetapi mereka lucu dan menarik
sangat menyenangkan untuk dipelajari dan menurut saya ini bagus untuk mempelajari buku matematika untuk naik
level berikutnya jadilah buku matematika sungguhan ini adalah beberapa buku matematika terbaik untuk pembelajaran mesin
(machine learning) buku elektronik (ebook) oleh dyson roth dan rekannya adalah sumber daya komprehensif yang
bagus dan tersedia gratis untuk pribadi penggunaannya mencakup topik-topik utama yang telah kita diskusikan seperti
kalkulus aljabar linier probabilitas (probability) dan pengoptimalan dengan fokus pada bagaimana konsep ini diterapkan pada
pembelajaran mesin (machine learning) algoritma (algorithm)s itu relatif ramah bagi pemula dan umumnya dianggap sebagai
salah satu buku terbaik untuk belajar materi statistik praktis berikutnya untuk ilmuwan data (data scientist)s
adalah sumber daya lain yang sangat disukai sekarang sebelum kita benar-benar mulai belajar, menurutku penting
sekali meluangkan sedikit waktu memikirkan secara mendalam mengapa Anda ingin melakukan ini secara

pribadi, saya
menemukannya jika saya sedang belajar Hanya karena aku merasa harus melakukannya atau karena ini adalah
tugas yang sewenang-wenang, perhatianku mudah teralihkan dan saya sebenarnya tidak menyimpan banyak informasi, melainkan saya
mencoba menyambungkannya semacam motivasi yang lebih dalam jika memungkinkan secara pribadi saat ini saya memiliki
motivasi yang sangat mendasar motivasi saya ingin melakukan pekerjaan saya dengan baik karena saya ingin
mendapatkan banyak uang sehingga saya dapat mengambil peduli pada semua orang yang kucintai, aku punya
sumber daya hebat untuk dipelajari, yang bahkan tidak ada tersedia beberapa tahun yang lalu dan
saya memiliki opporta (port)unity untuk mendorong diri saya sendiri dan memastikannya semua orang yang saya
cintai aman dan dirawat saat ini dan di masa depan, jadi ini bukan untuk memberikan
tekanan ekstra pada diri saya sendiri tetapi sebenarnya hanya cara yang berhasil bagi saya untuk
merasa bersemangat karena saya memiliki ini opporta (port)unity untuk belajar dan tumbuh dan semoga membantu
orang lain sepanjang motivasi Anda benar-benar berbeda tetapi apa pun itu, cobalah untuk mengaitkan pekerjaan
ini dengan tujuan yang lebih besar dalam hal pengoptimalan waktu belajar Anda saya menemukan bahwa
salah satu metode yang paling efektif adalah menulis catatan dengan kata-kata saya sendiri jangan hanya
menyalin definisi atau rumus luangkan waktu untuk meringkas konsep seolah-olah Anda sedang menjelaskannya kepada orang
lain atau sebenarnya kepada orang lain di masa depan misalnya jika Anda sedang belajar tentang
turunan, Anda mungkin akan menulis ukuran turunan bagaimana suatu fungsi berubah jika diberi sedikit dorongan
pada masukannya, hal ini memaksa Anda untuk aktif proses (process) materi yang berkaitan dengan rumus
matematika jangan hanya dilihat saja, aku tahu itu mudah dilakukan, sangat membantu untuk menerjemahkannya ke
dalam bahasa Inggris biasa atau bahasa lisan apa pun yang Anda gunakan lebih suka untuk
instans (instance) ambil persamaan y sama dengan mx ditambah b Anda dapat menggambarkan m sebagai
kemiringan yang menunjukkan kecuraman garis dan b sebagai kemiringan yang menunjukkan kecuraman

garis dan b

sebagai titik di mana garis melintasi sumbu y sehingga rumus akhirnya mungkin adalah nilai y

keluaran (output) ditentukan dengan mengambil kemiringan m mengalikannya dengan x masukan lalu menambahkan b permulaan

titik di mana garis memotong sumbu y, Anda bahkan dapat menggunakan catatan Anda seperti bcatatan

(log) pribadi Menulis postingan singkat tentang apa yang telah Anda pelajari adalah cara yang sangat

tepat untuk memperjelas pemahaman Anda memahami dan mengajar orang lain bahkan jika tidak ada yang

membacanya akan memperkuat materi dalam pikiran Anda sendiri ditambah membagikan postingan Anda di media atau

tautan (link)edin tidak hanya berpotensi membantu orang lain tetapi juga memungkinkan Anda membuat semacam porta

(port)folio yang menampilkan perjalanan belajar Anda, juga percayalah kepada saya saat itu intertampilan (view) kali

Anda akan sangat senang karena Anda memiliki catatan ini saya menggunakan studi saya sendiri mencatat

sepanjang waktu, nasihat berikutnya yang saya miliki mungkin tidak terlalu menyenangkan tetapi saya juga merekomendasikannya

tidak hanya menggunakan satu sumber daya secara pribadi saya telah meraih banyak keberhasilan dengan mengikuti

banyak kursus berbeda dan hanya mengumpulkan semua catatanku pada awalnya, lalu aku akan menulis bcatatan

(log) seolah-olah aku baru saja membicarakan hal itu merangkum semua pembelajaran saya, ada beberapa keuntungantag

(tag)es untuk pendekatan ini pengulangan pertama membantu Anda mempertahankan sesuatu jika saya melihat konsep berganda

kali dijelaskan dari berbagai sudut dan lebih mungkin untuk benar-benar memahami apa yang terjadi dan

reanggota (member) selama lebih dari satu hari ditambah saya tidak hanya melihat informasi yang disajikan

bagi saya berkali-kali saya menuliskan konsep dengan kata-kata saya sendiri beberapa kali termasuk terakhir kali

saya menyatukan semuanya dan bersiap untuk berbagi dengan orang lain, jadi saya harus melakukannya sangat

yakin bahwa saya benar-benar mendapatkannya pada akhirnya jika saya akan mempublikasikannya ke publik izinkan saya

menunjukkan kepada Anda sebuah contoh. Berikut adalah tiga penjelasan berbeda untuk konsep peringkat

dan aljabar
linier dimulai dengan tiga biru satu coklat Anda dapat melihat intuisi visual (visual) ini sedikit
lebih rumit daripada yang baru saja saya tunjukkan tetapi Anda tahu ada sedikit detail yang
kurang lalu kita bisa melihat bagaimana peringkat disajikan di deeplearning.ai yang memiliki beberapa detail lebih
lanjut masih disajikan dengan cara yang sangat sederhana akhirnya kita bisa lanjut ke matematika untuk
pembelajaran mesin (machine learning) buku elektronik (ebook) untuk mendapatkan definisi yang lebih formal dengan detail
tambahan pada setiap build ini di sisi lain dan akhirnya menambah simpanan pengetahuan saya setelah
Anda membangun fondasi itu dan mencapai tingkat matematika di mana Anda benar-benar dapat menggunakannya untuk
hal-hal yang sangat saya rekomendasikan untuk pengkodean konsep dari awal jika Anda dapat membuat kode
penurunan gradien (gradient descent) atau regresi logistik (logistic regression) hanya dengan menggunakan numpy kamu memulai
dengan sangat baik, sementara aku tahu saat ini kamu juga begitu sangat bersemangat untuk mulai
belajar matematika. Saya ingin kembali ke importa (port)ant fakta bahwa jika Anda seorang pemula mencoba
untuk mendapatkan pekerjaan pertama Anda menurut saya matematika tidak seharusnya menjadi hal pertama yang Anda
prioritaskan, sangat kecil kemungkinannya kemampuan matematika Anda akan tercapai hal yang memberi Anda pekerjaan sebagai
insinyur ilmuwan data (data scientist) atau pembelajaran mesin (machine learning) malah diprioritaskan mendapatkan pengalaman langsung
dengan mengerjakan proyek (project)s dan benar-benar membangun perusahaan jauh lebih banyak lagi tertarik untuk melihat
apa yang dapat Anda lakukan dengan alat (tool)s dan pengetahuan yang sudah Anda miliki daripada
bagaimana caranya banyak rumus yang Anda hafal saat menghadapi tantangan dalam pekerjaan Anda, tentu saja
Anda akan begitu matematika di balik matematika algoritma (algorithm)s reanggota (member) adalah alat (tool) untuk
membantu Anda sukses dan tidak boleh menjadi hambatan untuk memulai sekarang jika Anda penasaran tentang
apa lagi yang perlu Anda ketahui untuk menjadi seorang data ilmuwan saya memiliki video (video)

yang sangat komprehensif yang memaparkan keseluruhan peta jalan dari yang absolut pemula untuk pekerjaan pertama

Anda termasuk keterampilan untuk mempelajari proyek (project) contoh cara mendapatkan pengalaman mempersiapkan intertampilan (view)s dan

lebih pasti lihat berikutnya